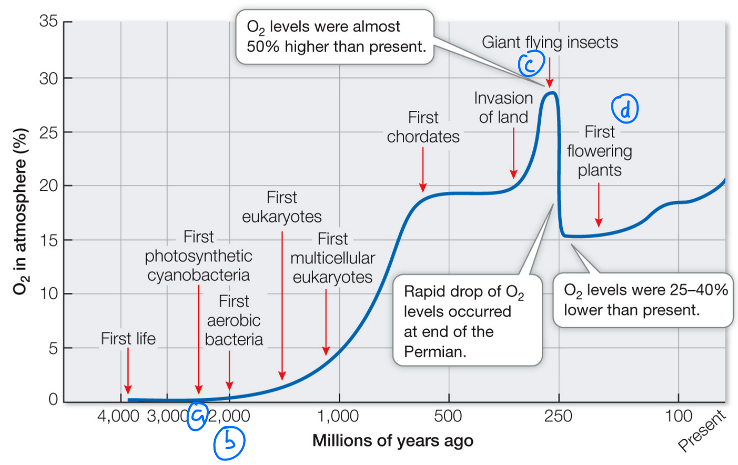
Study Guide for The Anthropocene and Conservation

This study guide provides examples for the topics that you might see on the final exam over the Anthropocene and Conservation lectures. This is NOT an exhaustive list of all content that might be covered on the exam. Use your notes from lecture, the lecture slides and video on Canvas, and your homework questions to review all the topics that have been covered.

1. Put the events below in the order in which they occurred in Earth's history.
   1 a. The origin of life on Earth
   2 b. The evolution of photosynthesis
   3 c. The colonization of land by early tetrapods
   4 d. The development of fossil fuels (oil, etc)
   5 e. Mass extinction

2. The diagram below shows the O_2 levels in the atmosphere throughout Earth's history. Important biological events are labeled.



   a. Photosynthesis evolves after the first evidence of life. If they were not photosynthetic, how did the first organisms obtain sugar for respiration?

   Through Chemosynthesis: (Carbon compounds break down → energy.

   b. The first evidence of aerobic respiration occurs about 2 billion years ago. Why did aerobic respiration arise after photosynthesis? If the first living organisms were not aerobic, how were they able to perform respiration?

   Huge availability of oxygen. They used to do anaerobic respiration.

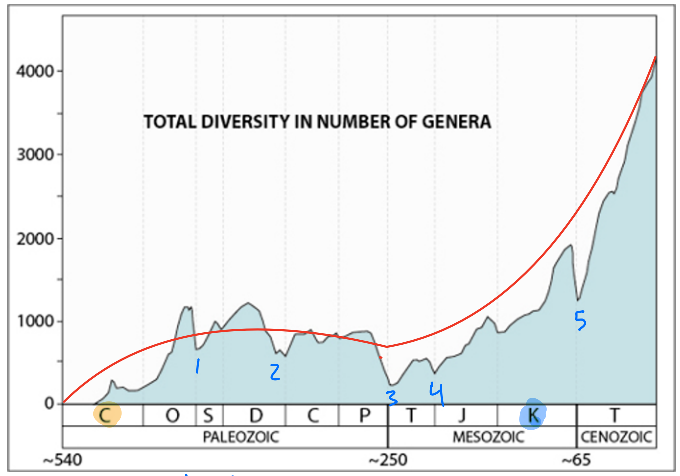
   c. Why do we see an increase in O2 levels following the invasion of land? What was the terrestrial landscape like at this time?

   Carboniferous: Massive forest / plant life on land increasing O_2 production.

   d. Why do O2 levels drop so significantly ~250 mya? Why do we not see similar drops 444 mya, 359 mya, 201 mya, or 65 mya?

d) Permian extinction event, 90% species wiped

3. The graph below shows the total number of genera (y axis) on Earth from ~540 million years ago to today (x axis).



Explosion = animal diversity exploded! Period of predation: Forced evo pressures: claws, jaws, armor, spines, movement.

a. The largest increase in genera is seen in the recent past, since the end of the Cretaceous Period. Why then is the Cambrian Period considered an 'explosion' of life?

b. What is the overall pattern in the figure? What does this indicate about the resilience of life on Earth? Upward, almost exponential

This pattern reflects resilience of life and process of rapid adaptation: Even after extinction events, biodiversity always trends up. Surviving species always repopulate + diversify again.

"Life Finds a Way" - Ian Malcolm

4. Describe how the diversification of angiosperms in the Paleogene influenced the terrestrial environment and the diversification of other taxonomic groups. Angio = Flowering plants (nectar and fruited seeds)

Increased transpiration, photosyn. Also influenced Nitrogen Cycle w/ Rhizobia bacteria.

↳ Easier to spread seeds pollen, etc.

5. What is the Anthropocene?

Not official geo time period (yet). Term for era when human activities became the main force shaping Earth's env.

Debate start dates: Indust Rev: 1750s, Homo sapiens: 300K years ago, Original Agri: 10K years ago

6. Give an example of each of the following responses to environmental change driven by human activity (Yellow Birds) (Blossoms) (HWY 101 + Bob cats)

a. Move: Climate change, Phenology, Physical Barriers
b. Acclimate: Physiology or behavior changes (Urban activity ⟷ nocturnal animal behavior)
c. Adapt: Rural vs Urban Lizards in PR limb length + lamellae tips (Urban Evo Pressure)
d. Die: If organisms cannot do any ↑ = death. White Rhino example

7. Compare and contrast the California Species of Special Concern, Endangered Species List, CITES-listed species, and the IUCN Red List. CA SSC: California list (herps, mammals, fish). IUCN Red List: Global assesment that rank species risk (vulnerable vs endangered); ESA w/ more threat categories

ESA: Federal (All taxo groups) managed by Fish/Wildlife Service. CITES: Int'l agreement; No habitat/pop protection.

8. What is the extinction vortex? What happens to a population that is in the vortex?

When a pop becomes small and problems keep reinforcing each other. Once a pop enters the cycle, it keeps declining towards extinction.

↳ (Problems: Inbreeding, low genetic div, random shrinking events

9. Describe an example of how each of the factors below threatens biodiversity

e. Habitat loss and alteration: Directly reduces pop size, increasing fragmentation, worsening the decline
f. Climate change: Can shift geo location of suitable hab to areas the species may not have access

10. What is the primary goal of California's 30x30 initiative? Are we on track to meet this goal?

Protect 30% of CA's area/coastal waters by 2030. Current = ~26% land, ~21% coastal. Possible with current trends

11. Fragmentation of habitat is a major problem for conservation. What strategies have been used to reduce fragmentation of remaining habitat? Phrased another way, how do conservation land managers increase connectivity between habitats?

Design Lands: Few, Larger protected areas within close proximity with protected corridors (paths) between; should limit area edges (round) and be surrounded by buffer zones. (Monarch Butterflies example)

Wildlife Corridors: Human made paths to help organisms impacted (crab bridges, bear tunnels, etc)

Remove human constructed barriers like dams (Elwha/Glines Canyon Dam for salmon revival)